

M1.5: Centres of Mass

1. Definition

For a system of particles with masses m_i at positions $\mathbf{r}_i$, the **centre of mass** is

$$\bar{\mathbf{r}} = \frac{\sum_i m_i \mathbf{r}_i}{\sum_i m_i}.$$

In uniform gravity, CoM coincides with the centre of gravity.

2. Uniform Laminae (2D)

For a region under $y = f(x)$ between $x = a, x = b$:

$$\bar{x} = \frac{\int_a^b x f(x) dx}{\int_a^b f(x) dx}, \quad \bar{y} = \frac{\frac{1}{2} \int_a^b [f(x)]^2 dx}{\int_a^b f(x) dx}$$

Standard Lamina CoMs

Shape	CoM position
Rectangle	Intersection of diagonals
Uniform triangle	Centroid: average of vertices
Semicircle radius r	$4r/(3\pi)$ from diameter

Circular sector, half-angle α	$2r \sin \alpha / (3\alpha)$ from centre
Circular arc, half-angle α	$r \sin \alpha / \alpha$ from centre

3. Solids of Revolution

About x -axis, $y = f(x)$, $a \leq x \leq b$:

$$\bar{x} = \frac{\int x[f(x)]^2 dx}{\int [f(x)]^2 dx}$$

Standard Solid CoMs (distance from base)

Solid	CoM
Hemisphere, radius r	$3r/8$
Cone apex up, height h	$h/4$
Pyramid	$h/4$
Hemispherical shell radius r	$r/2$
Conical shell height h	$h/3$

4. Composite Bodies

$$\bar{\mathbf{r}} = \frac{\sum m_i \bar{\mathbf{r}}_i}{\sum m_i}$$

Use **signed masses** (subtract) for holes.

Square lamina side 6 with circular hole radius 1 centred at (2,2). CoM.

Full square: area 36, CoM (3,3). Hole: area π , CoM (2,2). Remaining area $36 - \pi$. $\bar{x} = (108 -$

$2\pi)/(36 - \pi)$. Similarly $\bar{y}$.

5. Equilibrium and Toppling

A body on a horizontal surface is in equilibrium if the vertical through CoM falls within the base of support.

Toppling on Incline

Body (base-width $2a$, CoM height h) topples when $\tan \theta > a/h$, where θ is the inclination. Sliding occurs first if $\mu < a/h$.

Suspended Freely

CoM hangs directly below the suspension point.

6. Exam Strategy

- Use symmetry: CoM lies on every axis of symmetry.
- Composite: table of {part, mass (or area), $\bar{x}_i$, $\bar{y}_i$, $m_i\bar{x}_i$, $m_i\bar{y}_i$ }.
- Drawing the body tilted at the critical angle helps with toppling analysis.
- CoG and CoM coincide in uniform gravity — interchange freely.